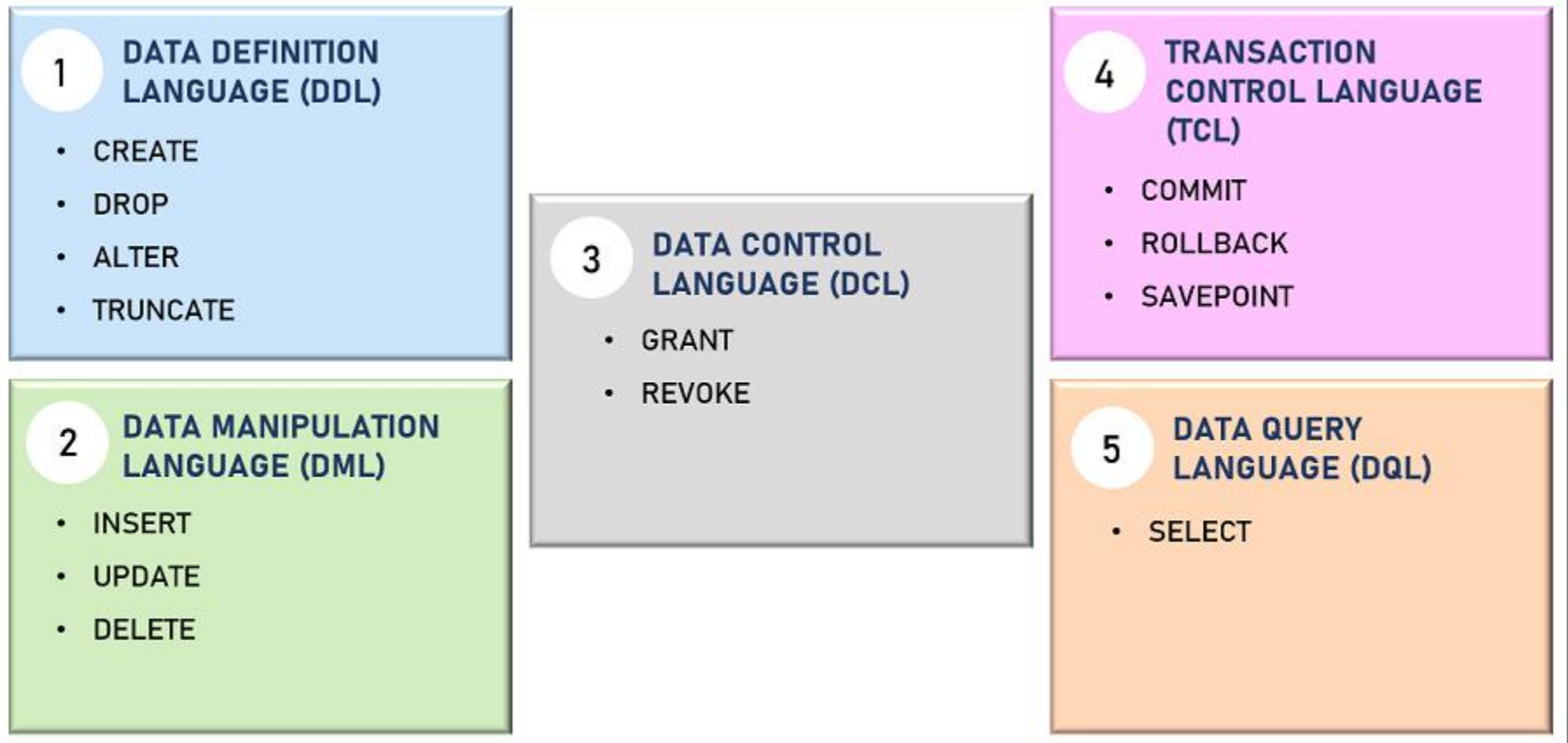


Lab 3

Introduction to Databases



SQL Commands



Common SQL Functions

- **Aggregate Functions:** COUNT(), SUM(), AVG(), MIN(), MAX()
- **String Functions:** UPPER(), LOWER(), CONCAT(), SUBSTRING()
- **Date Functions:** NOW(), CURDATE(), DATEDIFF()

SQL command exercises

1. Create a Table: Create a table STUDENTS with the following columns:

- ID → NUMBER (Primary Key)
- NAME → VARCHAR2(50)
- AGE → NUMBER
- GPA → NUMBER(3,2)
- DEPARTMENT → VARCHAR2(30)

2. Insert Data into a Table: Insert the following students into the table:

ID	Name	Age	GPA	DEPARTMENT
1	Ali	20	3.80	IS
2	Sara	22	3.20	IT
3	Rami	21	3.70	IS

- 3. Update Data into a Table:** Update GPA of student with ID = 1 to 3.90
- 4. Delete Data from a Table:** Delete student with ID = 3
- 5. ALTER TABLE :** Add a new column EMAIL VARCHAR2(100)
- 6. Drop:** Drop the EMAIL column
- 7. TRUNCATE:** Remove all data from STUDENTS table
- 8. Check data before and after TRUNCATE**
SELECT * FROM STUDENTS;

Home work

1. Create a table COURSES:

COURSE_ID NUMBER (PK)
COURSE_NAME VARCHAR2(50)
HOURS NUMBER

2. Insert 2 courses.

3. Add column PRICE NUMBER(6,2)

4. Update PRICE for one course.

5. Delete one course.

6. Truncate table.

7. Drop table.